Runway Re-Declaration Tool:
User Guide – Group 29.

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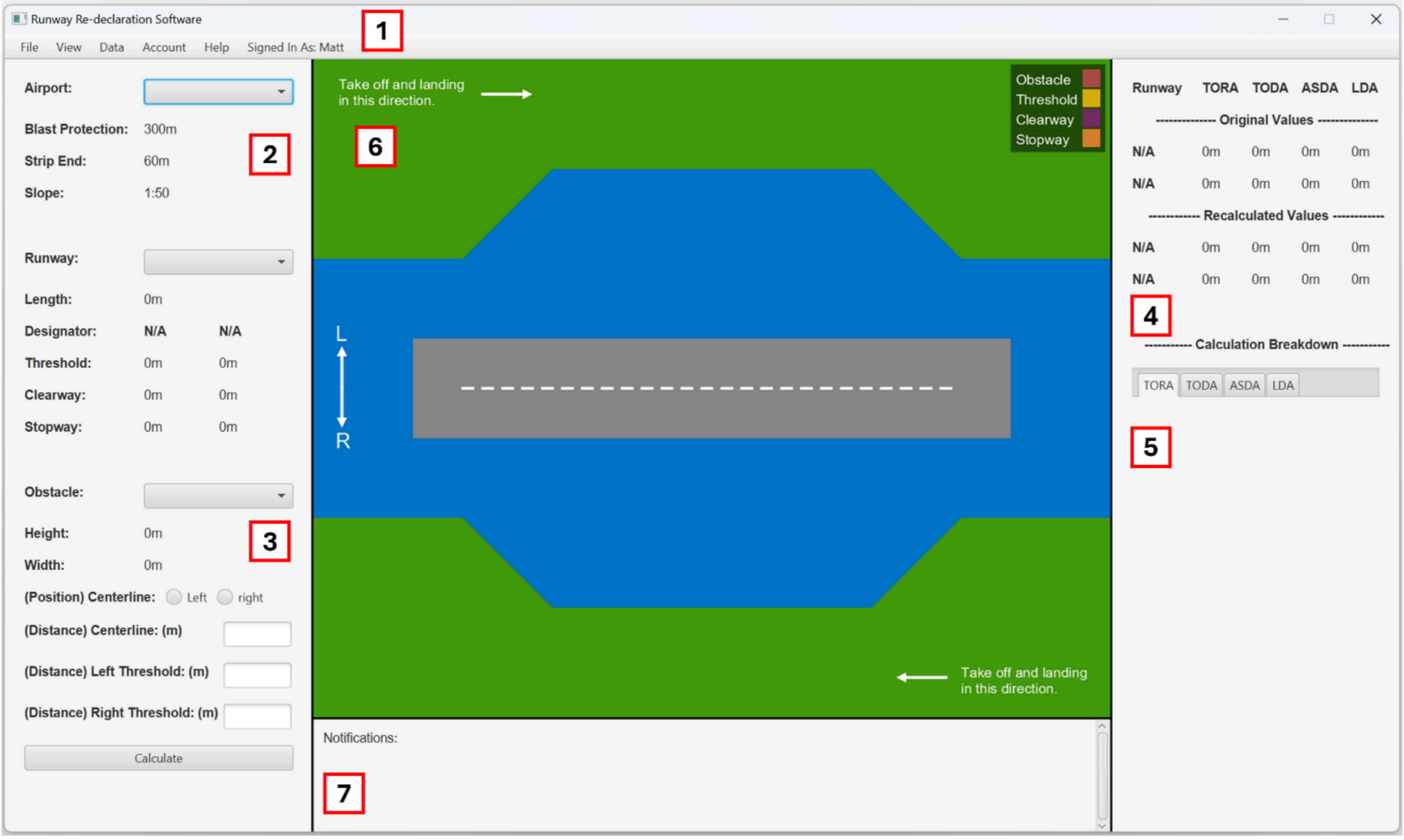


Section One: Main View.

1.1: Menu Bar.

The menu bar at the top of the screen provides access to features that cannot be accessed through the main view, these are outlined below but will be explained in greater detail in later sections:

File: Importing XML, Exporting to XML, PDF, and TXT. View: Switch between top-down and side-on views, runway orientations, also provides various accessibility features. Data: Manually add and edit airport, runway and obstacle parameters. Account Manage account information, allows admins to manage user accounts.

1.2: Runway Input.

This section can be used to select an airport and runway for use in calculations using dropdown boxes. When a runway is selected, its length is displayed, as well as the designation, threshold, clearway and stopway of each of its logical runways.

1.3: Obstacle Input.

This section can be used to select an obstacle for use in calculations using a dropdown box. When an obstacle is selected, its height and width are displayed. Input boxes can then be used to select the obstacle's Distance from the left and right runway thresholds, as well as the distance and direction from the runway's centreline.

The Distance from left and right thresholds signify the distance each runway threshold and by extension the length of the obstacle. For instance, if there was a 100m long obstacle 500m from the left threshold of a 3000m runway the parameters should be entered as follows: Distance from Left Threshold: 500. Distance from Right Threshold: 2400.

1.4: Calculation Results.

This section is a table that is regenerated after each set of calculations are run. It shows both the original and recalculated values for each logical runway of the selected runway. The values displayed are as follows: TORA: Take Off Runway Available. TODA: Take Off Distance Available. ASDA: Accelerate Stop Distance Available. LDA: Landing Distance Available.

1.5: Calculation Breakdowns.

This section shows a breakdown of all the calculations carried out by the software. The user has the option to select between four tabs, each corresponding to one of the recalculated parameters. Each breakdown includes the formula used, the values of each relevant parameter, and the calculation carried out for each logical runway.

1.6: Visualisation.

The section in the middle of the main window shows a visualisation of the selected runway, with annotated thresholds, clearways, stopways and recalculated runway parameters. Each arrow is labelled with its corresponding value, and a key is present to show the meaning of each colour. By default, the visualisation shows a top-down view of the runway, although the user can switch between top-down and side-on views using the 'View' menu. This menu can also be used to switch the orientation of the runway between horizontal (Default) and oriented by heading.

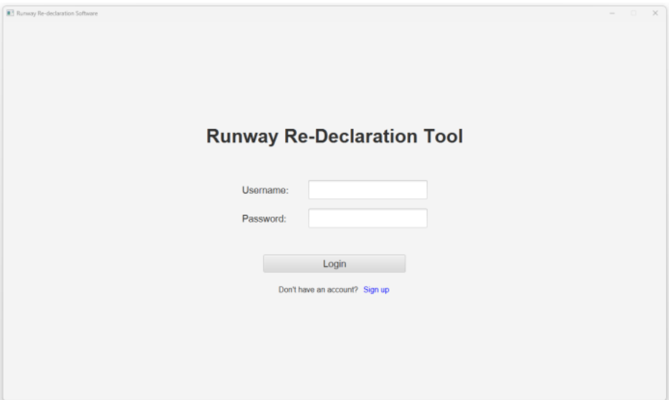
1.7: Notifications.

The notifications section provides a record of all the actions taken by the currently signed in user, as well as the timestamps at which they were performed. Records include importing and exporting data, running calculations, and many other actions.

Section Two: Authentication.

When the application is opened, the user is required to either Log-In or Sign-Up to access the main features of the application.

2.1: Log In.



The *Log-In* page allows users to log into an existing account by entering a username and password into two text boxes. After details are submitted, they are securely compared with the correct details and if they match, access to the application is granted

2.2: Sign Up.

The *Sign-Up* menu allows users to create and log into a new account by entering details into two text boxes. New usernames must be unique (Distinct from all other users) and for security reasons, passwords must be longer than 8 characters. Once a valid set of details have been submitted, a new account is created and access to the application is granted.

Section Three: Data Management.

3.1: Importing Via XML.

The primary method to configure application data is importing an XML document (Via *File* > *Import XML*), both airports and obstacles can be inputted this way. The required format is demonstrated in Appendix A.

To import an XML document, select *File* > *Import XML*. A prompt will be displayed allowing an XML file to be selected, select the relevant file and then Open.

3.2: Manual Data Configuration.

Airports, runways and obstacles can also be added and edited using the *Data* menu. Selecting the relevant option when using this menu will open a pop-up window to add / edit the relevant parameters.

The information required for an airport is: Name, Code and Runways. For each runway a length must be specified, along with values for TORA, TODA, ASDA, LDA, Threshold, Clearway, and Stopway for each of its two logical runways. For obstacles, a Name, Width and Height must be specified. After the information has been entered, the *add* or *edit* button should be selected and if the parameters are valid, the data will be added to the system.

Section Four: Exporting Data.

The application supports exporting to several different file formats, these can be viewed in the *File* tab. To export, select the relevant option from the dropdown menu, select the folder location in the popup, then enter a file name when prompted.

4.1: Exporting Data.

Selecting *File* > *Import Data* will generate an XML document containing the currently stored Airports and Obstacles, along with their parameters. The generated XML document will be of the format described in Appendix A, such that an exported XML document could be correctly imported by another instance of the application.

4.2: Exporting Calculations.

Selecting *File* > *Export Calculations* will generate a PDF report containing the contents of the most recently completed calculation (Note that a report cannot be generated if no calculations have been run). Specifically, each report contains Airport and Obstacle information, the original and recalculated runway parameters, calculation breakdowns, as well as top-down and side-on visualisations of the runway.

4.3: Exporting Notifications.

Selecting *File > Export Notifications* will generate a text file (.txt) containing a list of performed actions by the current user, along with the timestamps (Date and time) that they were performed.

Section Five: Accessibility.

Along with visualisation customisation options, the *view* menu provides several accessibility options. These are discussed below.

5.1: High Contrast Mode.

Selecting *View > High Contrast Mode* will switch the visualisation to a high contrast colour scheme. The colours used have been carefully selected from a specialised palette to maximise legibility.

5.2: Dark Mode.

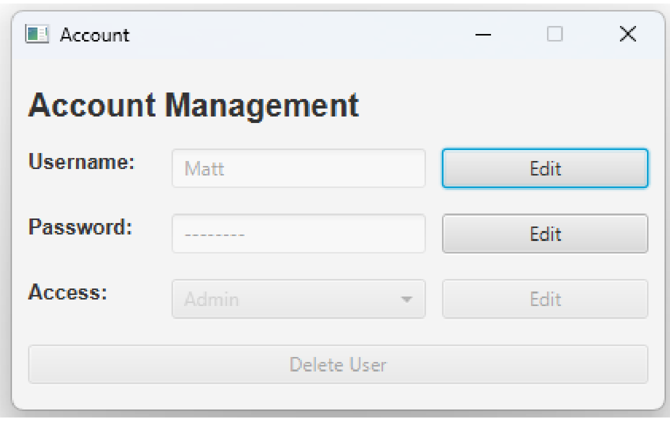
Selecting *View > Dark Mode* will enable dark mode. Specifically, a dark background with light text will be applied to all existing or newly created windows.

Section Six: User Management.

There are three levels of access to the system that can be assigned to an account (Admin, Staff, Guest), the permissions of each role are shown below.

	Guest.	Staff.	Admin.
Running Calculations.			
Data Management.			
View Settings.			
Accessibility Settings.			
Exporting Data.			
Account Management.			
User Management.			

5.1: Account.



Selecting *User > Account* will open the account management window. This can be used to change the current user's username and password. To change either of these, next to the relevant area select edit, enter the new data, and select submit. All users are unable to change their access level or delete their account, although these actions can be performed on other users by an account with admin access.

5.2: User Management.

If a user is logged into an account with admin access, selecting *User > User Management* will open the user management window. This window provides a scrollable list of all user accounts (Excluding the one currently logged in), with an option to view the details of each account.

Selecting view details will open an account management window relating to the selected account. This will provide the option to edit the account's username, password, and access level, as well as delete the account. For information on how to do this, see section 5.1.

Section Seven: Frequently Asked Questions.

Q: Why does the distance of an obstacle from each end of the runway need to be rather than the length?
A: The distance from each end of the runway is required to determine which direction should be used for take-off and landing.

Q: What is the difference between orientations *Horizontal* and *Oriented by Heading*.
A: *Horizontal* displays the runway relative to the display window. *Oriented by Heading* displays the runway relative to its real-world orientation such that the top of the display points north.

Q: I've got an error saying that the obstacle is too far away from the runway, what does this mean?
A: No re-declaration is required if an obstacle is further than 75m from the centreline, or 60m from the runway.

Q: I've got an error saying that a visualisation cannot be generated, what does this mean?
A: One of the re-calculated runway parameters (TORA, TODA, ASDA, LDA) is less than zero and so would cause an error when generating a visualisation.

Section Eight: Known Issues.

If the application is opened after not being used for several days, when attempting to log in or sign up, the user may be shown an error stating that the program cannot connect to the database. In order to fix this, wait for around one minute, then attempt to log in or sign up again

Appendices.

Appendix A: Valid XML Import and Output Format.


```
<data>
    <airports>
        <airport>
            <name></name>
            <code></code>
            <runways>
                <runway>
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    </airports>
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            <name>Airbus </name>
            <width></width>
            <height></height>
        </obstacle>
    </obstacles>
</data>
```